

Project Development Phase

Coding & Solution

Date	10/07/2026
Name	Karuru Arjun
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Coding & Solution:

This section evaluates the implementation of the **Voice-Based Concept Understanding Analyser**. The solution is developed using Python and machine learning libraries to predict whether a credit card application should be approved based on applicant details. The project follows clean coding practices, modular programming, proper documentation, and error handling to ensure maintainability and scalability.

Instructions:

1. The project code is modular, well-structured, and follows Python coding standards.
2. The solution successfully predicts credit card approval using trained machine learning models.
3. All required source files, trained model files, datasets, and dependency files are included.
4. The project can be maintained using a Git repository for version control.

Solution Summary:

Field	Details
Repository Link / URL	https://github.com/23F21A3110/VBCUA
Programming Language(s)	Python 3.x
Framework(s) Used	Streamlit
AI Libraries	OpenAI Whisper, Sentence-Transformers
Audio Processing	Librosa, NumPy
Visualization	Matplotlib, Plotly.

Code Quality Checklist:

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (<i>if using GitHub</i>)

Additional Notes / Comments:

The VBCUA project adopts a modular architecture by separating speech transcription, semantic evaluation, audio feature extraction, scoring, report generation, and visualization into independent modules. The implementation follows Python coding standards (PEP 8) to improve readability and maintainability. AI models including OpenAI Whisper and Sentence-BERT are integrated for intelligent speech understanding, while Librosa is used for extracting speech fluency features. The application is designed to be scalable and can be extended with cloud deployment, user authentication, multilingual speech analysis, and database integration for long-term progress tracking.